

Pranav Balaji Kapratwar

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Summary

Final-year B.Tech Computer Science (Data Science) student with hands-on experience building **Machine Learning and Generative AI pipelines** covering data preprocessing, model training, evaluation, and inference. Skilled in Python, NumPy, Pandas, Flask, **FastAPI**, LangChain, FAISS, Grok API, TensorFlow, Streamlit, and SQL. Proven experience developing **RAG-based and NLP workflows**, including text processing, data cleaning, parsing, **chunking**, and **embeddings**, along with **backend API development** to expose ML/AI functionalities. Comfortable writing clean, modular, and testable Python code, integrating AI models into applications, and supporting deployment activities. Exposure to **cloud platforms (AWS)**, Git, and collaborative software development practices, with a strong foundation in Data Structures, OOP, DBMS, and problem-solving. Quick learner currently expanding into Computer Vision and YOLO-based object detection.

Skills

- **Languages:** Python, SQL
- **Backend / APIs:** Flask, FastAPI, REST API development, Backend API Integration
- **Frontend:** HTML, CSS, Streamlit
- **ML / AI:** Machine Learning, Deep Learning, NLP, Model Evaluation, Scikit-Learn, TensorFlow, NLTK, Pandas, NumPy, Data Preprocessing, Model Training & Inference
- **Generative AI:** Large Language Models (LLMs), Prompt Engineering, Generative AI, Retrieval-Augmented Generation (RAG), Text Chunking, Embeddings, LangChain
- **Also Working With:** PyTorch, ANN, CNN, RNN, Computer Vision, YOLO, Hugging Face Transformers
- **Database:** MySQL, SQLite, DBMS
- **Cloud:** AWS
- **Mathematics:** Statistics, Probability, Linear Algebra
- **Core CS:** OOP, DSA, OS Basics, DBMS
- **ML Practices:** Feature Engineering, Hyperparameter Tuning, Model Deployment, Experimentation, Cross-Validation, Clean & Modular Code Practices
- **Tools:** GitHub, Git, VS Code, BentoML, Pickle, Docker
- **Soft Skills:** Communication, Leadership, Time Management, Presentation

Projects

StudyBot — AI-Powered RAG Study Assistant

2026

Python · Streamlit · LangChain · Grok API · FAISS · SQLite · NLP · Generative AI · RAG

github.com/Pranavkapratwar10/studybot

- Developed a secure AI-powered study assistant with user authentication, allowing students to upload multiple study materials and ask questions in natural language.
- Built an end-to-end **Retrieval-Augmented Generation (RAG) pipeline** using LangChain, a FAISS vector database, and the Grok Large Language Model (LLM) API, including **document parsing, chunking, and embeddings generation** to retrieve relevant content and generate context-aware answers.
- Engineered prompts and integrated **Generative AI API calls** to power five exam preparation modes including Viva, MCQs, Short Answers, Long Answers, and Revision Notes, along with intelligent fallback search when relevant content was unavailable.
- Designed an admin dashboard with SQLite-based user management, persistent chat history, document tracking, and system statistics for monitoring application usage.

TalentTrack — Campus & Corporate Hiring Platform

2025

Flask · HTML · CSS · SQLite · pdfplumber · Tesseract OCR · REST API · NLP

github.com/Pranavkapratwar10/Talentrack1

- Built a dual-panel web app using Flask and SQLite allowing admins to post jobs and students to register, upload resumes, and apply from one portal.
- Developed a resume parser using pdfplumber and Tesseract OCR to extract structured data from any PDF format including scanned files, involving **text preprocessing and data cleaning**.
- Designed an ATS ranking engine using **NLP-based similarity scoring** and feature engineering to rank applicants automatically.
- Implemented an email notification system for selection/rejection with reasons and automated vacancy cap enforcement via **REST API endpoints**.

Traconomy — Personal Finance Management System

2025

Python · Flask · HTML · CSS · SQLite · REST API

github.com/Pranavkapratwar10/traconomy

- Built a full-stack finance tracker with expense logging, savings goals, and custom categories using Flask and SQLite, exposed through **REST API** endpoints.
- Implemented downloadable Excel and PDF report generation for financial summaries.
- Integrated automated email alerts for goal completion and priority-based financial event reminders.
- Added EMI, SIP, FD, and Tax calculators alongside calendar-based monthly performance tracking.

AI-Powered Student Feedback Analyzer

Sept – Oct 2025

Python · Scikit-Learn · NLP · SVM · Random Forest · Streamlit

github.com/Pranavkapratwar10/my-project-student-feedback-analyzer

- Developed an **NLP and Machine Learning pipeline** processing multilingual feedback text, including data cleaning and preprocessing, achieving 92% classification accuracy while reducing manual review time.
- Designed a sensitive-issue detection module to flag harassment and discrimination in feedback data.
- Created an interactive Streamlit dashboard for trend visualization and automated report generation.
- Applied tuned SVM and Random Forest models, with **model evaluation and optimization** to maximize classification performance.
- Secured Runner-Up at Technovation 1.0 Hackathon among 100+ teams.

Internship

SaiKet Systems

Oct 2025 – Nov 2025

Machine Learning Intern

- Built and maintained an **end-to-end ML pipeline** (data preprocessing, feature engineering, training, evaluation, and inference) on an unstructured hotel management dataset using Python and Pandas, handling missing values, inconsistent formats, and noise.
- Performed feature engineering and exploratory data analysis (EDA) to uncover booking patterns, cancellation trends, and seasonal demand insights.
- Trained a Random Forest classifier for cancellation prediction; improved performance via hyperparameter tuning and cross-validation, following clean, modular, and testable coding practices with Git-based version control.
- Achieved high predictive performance after hyperparameter tuning and cross-validation, demonstrating strong model generalization and reliable pipeline optimization.

Education

B.Tech Computer Science (Data Science)	2023 – Present
Kolhapur Institute of Technology	CGPA: 8.2
Class XII — Maharashtra Board	2023
Shahu Maharaj College, Loha	Percentage: 71.5%
Class X — Maharashtra Board	2021
Nalge High School, Loha	Percentage: 92.2%

Achievements

- Runner-Up — Technovation 1.0 Hackathon among 100+ competing teams.
- Published dataset on Kaggle with 14K+ views and 3K+ downloads.
- Solved 130+ problems on LeetCode focusing on core DSA patterns.

Certifications

- 100 Days of Code: Complete Python Pro Bootcamp
 - Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp
- Oct 2024 – Jan 2025
- Jan 2026 – Mar 2026

Languages

English — Hindi — Marathi